KARNATAKA BOARD
MATHEMATICS
Class 10
2017

Time: 3hrs **M.M: 80**

General Instructions to the Candidate:
1. This Question Paper consists of 40 objective and subjective types of questions.
2. This question paper has been sealed by reverse jacket. You have to cut on the right side to open the paper at the time of commencement of the examination. Check whether all the pages of the question paper are intact.
3. Follow the instructions given against both the objective and subjective types of questions.
4. Figures in the right hand margin indicate maximum marks for the questions.
5. The maximum time to answer the paper is given at the top of the question paper. It includes 15 minutes for reading the question paper.

SECTION - A

I. Four alternatives are given for each of the following questions /incomplete statements. Only one of is correct or most appropriate. Choose the correct alterative and writhe the complete answer along with its letter of alphabet.

1. If a polynomial $p(x) = x^2 - 4$ is divided by a linear polynomial $(x - 2)$ then the remainder is
(A) 2 (B) – 2
(C) 0 (D) – 8.

Ans. 2

2. The sum and product of the roots of the equation $x^2 + 2x + 1 = 0$ are respectively,
(A) 2 and – 1
(B) – 2 and 1
(C) – 2 and – 1
(D) 1 and 2.

Ans. – 2 and 1

3. In a circle the angle between a radius and a tangent at non-centre end of the radius is
(A) 90°
(B) 180°
(C) 45°
(D) 360°.

Ans. 90°

4. The volume of a right circular cylinder whose circular base area is 154 sq.cm and height 10 cm is

(A) 15·40 c.c.
(B) 15400 c.c.
(C) 1·540 c.c.
(D) 1540 c.c.

Ans. 1540 c.c.

5. If tan θ = $\frac{1}{\sqrt{3}}$ and cos θ = $\frac{\sqrt{3}}{2}$ then the value of sin θ is

(A) $\sqrt{3}$

(B) $\frac{1}{2}$

(C) $\frac{2}{\sqrt{3}}$

(D) $\frac{3}{2}$

Ans. $\frac{1}{2}$

6. ($7 \times 11 \times 13 + 13$ **) is a / an**

(A) Composite number
(B) Prime number
(C) Irrational number
(D) Imaginary number.

Ans. Composite number

7. The sum of an infinite geometric series whose first term is *a* and common ratio is *r* is given by

(A) $S_\infty = \frac{1}{a-r}$

(B) $S_\infty = \frac{1}{r-a}$

(C) $S_\infty = \frac{a}{1-r}$

(D) $S_\infty = \frac{1-r}{a}$

Ans. $S_\infty = \frac{a}{1-r}$

8. Lateral surface area of the frustum of a cone is

(A) $\pi(r_2 - r_1)h$
(B) $\pi(r_1 + r_2)h$
(C) $\pi(r_1 - r_2)l$
(D) $\pi(r_1 + r_2)l$

Ans. $\pi(r_1 + r_2)l$

II. Answer the following:

9. If *U* = { 1, 2, 3, 4, 5, 6 } and *A* = { 2, 3, 4, 5 } then find *A'*.

Ans. . A' = U – A

A' = { 1, 2, 3, 4, 5, 6 } - { 2,3, 4, 5 }

∴ A' = {1, 6}

10. Write the relation between standard deviation of a set of scores and its variance.

Ans. The standard deviation of a set of scores is the square root of its variance. That means,

Standard deviation = √variance

11. In a sequence if $T_n = n^2 + 4$ then find T_2.

Ans.

$$T_n = n^2 + 4$$
$$\therefore T_2 = 2^2 + 4$$
$$= 4 + 4$$
$$= 8$$

12. A fair coin is tossed once. Find the probability that head occurs.

Ans. Total number of outcomes = {H, T} = 2

∴ Probability of occurring head = P(H) = $\frac{1}{2}$

13. State Pythagoras theorem.

Ans. In a right angled triangle, the square of the hypotenuse is equal to the sum of the square of the other two sides.

14. Write the general form of a quadratic polynomial.

Ans. Standard form of quadratic polynomial is $f(x) = ax^2 + bx + c$ where a ≠ 0, and a, b, c are real or complex numbers.

III. Answer the following:

15. Given A = { 1, 2, 3, 4 }, B = { 3, 4, 5, 6 } and C = { 6, 7 }. Verify that ($A \cap B$) $\cap C = A \cap$ ($B \cap C$).

Ans. Given,

A = { 1, 2, 3, 4 },
B = { 3, 4, 5, 6 } and
C = { 6, 7 }

Here, (A ∩B)= {3, 4}
∴ (A ∩B) ∩C = 0

Also,
($B \cap C$) = {6}
∴ $A \cap$ ($B \cap C$) = 0

Hence, ($A \cap B$) $\cap C = A \cap$ ($B \cap C$) verified.

16. Arithmetic mean between two numbers is 5 and Geometric mean between them is 4. Find the Harmonic mean between the numbers.

Ans.
Let a & b be the two numbers.
Then,

Arithmetic mean (A.M) = $\frac{a+b}{2}$

Geometric mean (G.M) = $\sqrt{ab}$

Harmonic mean (H.M) = $\frac{2ab}{a+b}$

Here,

(A.M) ×(H.M) = $\frac{a+b}{2} \times \frac{2ab}{a+b}$ = ab

∴(A.M) ×(H.M) = $(G..M)^2$ $[\because G.M = \sqrt{ab}]$

Now, substituting the given values, we get

$5\times(H.M)=(4)^2$

$\Rightarrow (H.M)=\dfrac{16}{5}$

$\therefore (H.M)=3.2$

Therefore, the harmonic progression between the numbers is 3.2

OR

In a harmonic progression third term and fifth terms are respectively 1 and $\dfrac{1}{-5}$. Find the tenth term.

Ans:

The third term of a harmonic progression is = $\dfrac{1}{a+2d}$

The fifth term of a harmonic progression is = $\dfrac{1}{a+4d}$

$\dfrac{1}{a+2d}=1$

$1=a+2d$**............................(1)**

$\dfrac{1}{a+4d}=\dfrac{1}{-5}$

$-5=a+4d$**...............................(2)**

Subtracting (1) from (2)

$-6=2d$

$d=-3$

From (1)

$1=a+2(-3)$

$1+6=a$

$a=7$

The tenth term of a harmonic progression is = $\dfrac{1}{a+9d}=\dfrac{1}{7-21}=\dfrac{1}{-14}$

17. Prove that $5-\sqrt{3}$ is an irrational number.

Ans: Ans: Let's assume $5-\sqrt{3}$ is a rational number

$$5-\sqrt{3}=\frac{a}{b}$$
$$\sqrt{3}=5-\frac{a}{b}$$
$$\sqrt{3}=\frac{5b}{b}-\frac{a}{b}$$
$$\sqrt{3}=\frac{5b-a}{b}$$

As we know that $\sqrt{3}$ is an irrational number so, $\frac{5b-a}{b}$ is also an irrational number. Therefore, our assumption is correct.

Hence, $5-\sqrt{3}$ is an irrational number.

18. Find the value of *n* if ${}^{n}P_{4}=5({}^{n}P_{3})$.

Ans:

$${}^{n}P_{4}=5({}^{n}P_{3})$$
$$\frac{n!}{(n-4)!}=5\left(\frac{n!}{(n-3)!}\right)$$
$$\frac{n!}{(n-4)!}\times\left(\frac{(n-3)!}{n!}\right)=5$$
$$\frac{n!}{(n-4)!}\times\frac{(n-3)(n-4)!}{n!}=5$$
$$n-3=5$$
$$n=8$$

19. If A is an event in a random experiment such that $P(A) : P(\bar{A}) = 5 : 11$, then find $P(A)$ and $P(\bar{A})$.

Ans:

$$P(A) : P(\overline{A}) = 5 : 11$$

$$\frac{P(A)}{P(\overline{A})} = \frac{5}{11}$$

$$\frac{P(A)}{1 - P(A)} = \frac{5}{11}$$

$$11P(A) = 5 - 5P(A)$$

$$16P(A) = 5$$

$$P(A) = \frac{5}{16}$$

$$P(\overline{A}) = 1 - \frac{5}{16} = \frac{11}{16}$$

20. What are like surds and unlike surds? Identify and write the set of like surds in the following groups:

a) $\left\{\sqrt{8}, \sqrt{12}, \sqrt{20}, \sqrt{54}\right\}$

b) $\left\{\sqrt{50}, \sqrt[3]{54}, \sqrt[4]{32}\right\}$

c) $\left\{\sqrt{8}, \sqrt{18}, \sqrt{32}, \sqrt{50}\right\}$.

Ans: Two or more surds are said to be similar or like surds if they have the same surd-factor or if they can be so reduced as to have the same surd-factor, else they are unlike surds.

Among the given groups, the like surds are:

$$\left\{\sqrt{50}, \sqrt[3]{54}, \sqrt[4]{32}\right\} = 5\sqrt{2}, 3\sqrt{2}, 2\sqrt{2}$$

$$\left\{\sqrt{8}, \sqrt{18}, \sqrt{32}, \sqrt{50}\right\} = 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, 5\sqrt{2}$$

21. Rationalise the denominator and simplify: $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$

Ans:

$$=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}\times\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$$
$$=\frac{\left(\sqrt{5}+\sqrt{3}\right)^2}{\left(\sqrt{5}\right)^2-\left(\sqrt{3}\right)^2}$$

$$=\frac{5+2\sqrt{15}+3}{5-3}$$
$$=\frac{8+2\sqrt{15}}{2}$$
$$=4+\sqrt{15}$$

22. A polynomial *p* (x) is divided by (2x – 1). The quotient and remainder obtained are ($7x^2 + x + 5$) and 4 respectively. Find *p* (x).

Ans: Dividend = Divisor X Quotient + Remainder

$$Dividend=\left[(2x-1)(7x^2+x+5)\right]+4$$
$$P(x)=\left[2x(7x^2+x+5)-1(7x^2+x+5)\right]+4$$
$$P(x)=\left[\left(14x^3+2x^2+10x\right)-(7x^2+x+5)\right]+4$$
$$P(x)=\left[14x^3+2x^2-7x^2+10x-x-5\right]+4$$
$$P(x)=14x^3-5x^2+9x-5+4$$
$$P(x)=14x^3-5x^2+9x-1$$

OR

Find the quotient and remainder using synthetic division.
($3x^3 - 2x^2 + 7x - 5$ **) ÷ (x + 3)**
Ans:

-3	3	-2	7	-5
		9	-21	42
	-3	7	-14	37

$$3x^2+7x+-14-\frac{37}{x-3}$$

23. Area of an equilateral triangle is given by $A = \frac{\sqrt{3}a^2}{4}$ where A is the area and a is the side. Find the perimeter of the triangle if $A = 16\sqrt{3}$ sq.cm.

Ans: $A = \frac{\sqrt{3}a^2}{4}$

$$16\sqrt{3} = \frac{\sqrt{3}a^2}{4}$$
$$16\sqrt{3} \times \frac{4}{\sqrt{3}} = a^2$$
$$a^2 = 64$$
$$a = 8$$

24. Show that the roots of the equation $x^2 - 2x + 3 = 0$ are imaginary.

Ans: We need to find the discriminant (d). If

d = 0, the equation has one real root

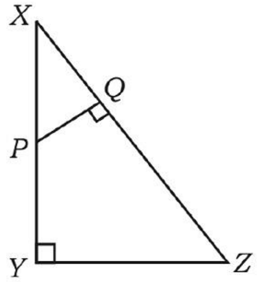
d < 0, the equation has two complex roots

d > 0, the equation has two real roots

$$d = b^2 - 4ac$$
$$d = (-2)^2 - 4(1)(3)$$
$$d = 4 - 12$$
$$d = -8$$

As d < 0, therefore, the given equation has two complex or imaginary roots

25. In ΔXYZ, P is any point on XY and $PQ \perp XZ$. If XP = 4 cm, XY = 16 cm and XZ = 24 cm, find XQ.



Ans: In ΔXYZ and ΔXPQ

$\angle XQP = \angle XYZ = 90°$

$\angle QXP = \angle YXZ$ (Common)

$\therefore \Delta XYZ \sim \Delta XPQ$

$$\frac{XP}{XY} = \frac{XQ}{XZ}$$

$$\frac{4}{16} = \frac{XQ}{24}$$

$$\frac{4 \times 24}{16} = XQ$$

$$XQ = 6cm$$

26. Show that $\dfrac{1-\tan^2 A}{1+\tan^2 A} = 2\cos^2 A - 1.$

Ans.

$$\frac{1-\tan^2 A}{1+\tan^2 A}$$

$$= \frac{1-\dfrac{\sin^2 A}{\cos^2 A}}{\sec^2 A}$$

$$= \frac{\dfrac{\cos^2 A - \sin^2 A}{\cos^2 A}}{\dfrac{1}{\cos^2 A}}$$

$$= \cos^2 A - \sin^2 A$$

$$= \cos^2 A - (1-\cos^2 A)$$

$$= \cos^2 A - 1 + \cos^2 A$$

$$= 2\cos^2 A - 1$$

27. Find the slope of the line joining the points (4, – 8) and (5, – 2).

Ans: Given, (4, – 8) = (x_1, y_1)

(5, – 2) = (x_2, y_2)

$$\therefore \text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$= \frac{-2-(-8)}{5-4}$

$= \frac{-2+8}{1}$

$= 6$

28. Find the co-ordinates of the mid-point of the line joining the points (2, 3) and (4, 7).

Ans. Given, (2, 3)= (x_1, y_1)

(4, 7)= (x_2, y_2)

We know, formula for Co-ordinates of mid-point = $\left\{\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right\}$

$= \left\{\frac{2+4}{2}, \frac{3+7}{2}\right\}$

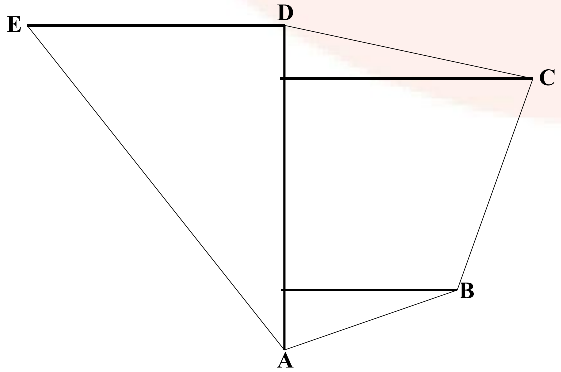
$= \left\{\frac{6}{2}, \frac{10}{2}\right\}$

$= (3,5)$

29. Draw a plan of a level ground using the information given below: [Scale: 20 metres = 1 cm]

	To D (in metres)	
80 to E	**150**	
	100	**70 to C**
	80	
	30	**40 to B**
	From A	

Ans:



30. Draw a circle of radius 3·5 cm and construct a chord of length 6 cm in it. Measure and write the distance between the centre and the chord.

Ans. Steps of construction:

1. Draw a circle of radius 3.5 cm marked as OA.
2. Draw a chord on the circle with length 6 cm as AB.
3. Draw a perpendicular bisector from the centre of the circle to the chord which bisects the chord into equal parts by bisector theorem.
4. Marked the length as OC.

Now, take triangle OAC,
By Pythagoras theorem,

$OA^2 = OC^2 + AC^2$

$\Rightarrow (3.5)^2 = OC^2 + 3^2$

$\Rightarrow 12.25 = OC^2 + 9$

$\Rightarrow OC^2 = 3.25$

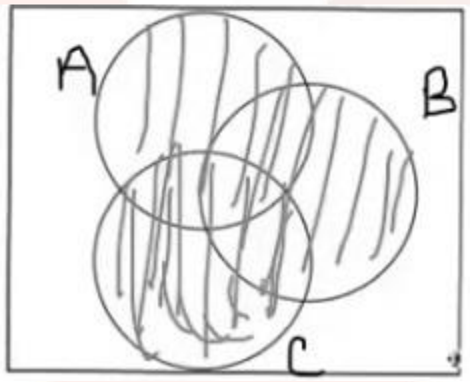
$\Rightarrow OC = \sqrt{3.25}$

$\therefore OC = 1.87$

Hence, the distance between the centre and the chord is 1.87 m.

31. *A*, *B* and *C* are the sets having some common elements in them. Draw Venn diagram to illustrate (*A* ∪ *B*) ∪ *C*.

Ans:



32. Find the sum of the series 1 + 2 + 4 + ... up to 10 terms.

Given series is in geometric progression.

First term (a) = 1

Common ratio (r) = $\frac{2}{1}$ = 2

$\therefore$ Sum of first 10 terms = $a\left(\dfrac{1-r^n}{1-r}\right)$

$$=1\left(\frac{1-2^{10}}{1-2}\right)$$

$$=1\left(\frac{1-1024}{-1}\right)$$

$$=1023$$

33. A survey which was conducted to study the various brands of soaps used by some people in a village is given below. Draw pie chart to represent the data.

Brand of soap	A	B	C
No. of people used	12	04	08

Ans:

34. Simplify: $\sqrt{18}+\sqrt{128}-\sqrt{50}$ **.**

Ans. $\sqrt{18}+\sqrt{128}-\sqrt{50}$

$= 3\sqrt{2}+8\sqrt{2}-5\sqrt{2}$

$=11\sqrt{2}-5\sqrt{2}$

$= 6\sqrt{2}$

35. If ${}^5C_r$ **= 10 and** ${}^5P_r$ **= 60 then find the value of *r*.**

Ans:

$${}^5P_r = \frac{5!}{(5-r)!}$$

$$60 = \frac{120}{(5-r)!}$$

$$(5-r)! = 2$$

$$2! = 2$$

$$5-r = 2$$

$$r = 3$$

To verify

$$^5C_3 = \frac{5!}{3!(5-3)!}$$

$$10 \stackrel{?}{=} \frac{120}{3!2!}$$

$$10 \stackrel{?}{=} \frac{120}{3\times2\times1\times2\times1}$$

$$10 \stackrel{?}{=} \frac{120}{12}$$

$$10 = 10$$

36. Solve the equation $x^2 - 7x + 12 = 0$ by using formula.

Ans. Given equation is $x^2 - 7x + 12 = 0$

Comparing it with $ax^2 + bx + c = 0$, we get

a =1 , b =-7 and c = 12

We know,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-7) \pm \sqrt{(-7)^2 - 4\times1\times12}}{2\times1}$$

$$= \frac{7 \pm \sqrt{49-48}}{2}$$

$$= \frac{7 \pm \sqrt{1}}{2}$$

$$= \frac{7 \pm 1}{2}$$

$$= \frac{7+1}{2} \text{ or } \frac{7-1}{2}$$

= 4 or 3

37. The corresponding altitudes of two similar triangles are 3 cm and 5 cm respectively. Find the ratio between their areas.

Ans. Given, altitudes of two similar triangles are 3 cm and 5 cm respectively.

We know,

$$\frac{\textit{Area} \text{ of one triangle}}{\textit{Area} \text{ of another triangle}} = \left(\frac{\textit{length} \text{ of one triangle}}{\textit{length} \text{ of another triangle}}\right)^2$$

$= \left(\frac{3}{5}\right)^2$

$= \frac{9}{25}$

Therefore, ratio of the given similar triangles is 9:25.

38. Find the remainder when *P* (x) is divided by

a) $(x - a)$
b) $(x + a)$

Ans.

When P(x) is divided by (x-a), the remainder is P(a).

When P(x) is divided by (x+a), the remainder is P(-a).

39. The height of a right circular cone is 4 cm and radius of its circular base is $\frac{21}{2}$ cm. Find the volume of it.

Ans. Given, radius (r) = $\frac{21}{2}$ cm

Height (h) = 4 cm

$\therefore$ Volume of the right circular cone = $\frac{1}{3}\pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times \left(\frac{21}{2}\right)^2 \times 4$$

$$= \frac{1}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times 4$$

$$= 22 \times 21$$

$$= 462 cm^3$$

40. In the figure AB, BC and AC are the tangents to circle of centre O. If AB = AC, show that BQ = CQ.

Ans. Given, AB, BC and AC are the tangents to circle at P, Q and R.

Using theorem that when two different tangents are drawn to the same circle at a common point, the distance between that point and the two points of tangency are the equal, we get

AP=AR...(1)

BP=BQ...(2)

CR=CQ...(3)

It is given that AB=AC

AP+BP=AR+CR

AP+BP=AP+CR (∵ From 1)

∴BP=CR

From eq (2) and (3), we get

BQ=QC

Hence proved.

IV Answer the following:

41. Everybody in a function shakes hand with everybody else. The total number of handshakes are 45. Find the number of persons in the function.

Ans: Let the number of people in the function be x

Then total number of handshakes are = $\frac{x(x-1)}{2}$

$\Rightarrow \frac{x(x-1)}{2} = 45$

$x(x-1) = 90$

$x^2 - x - 90 = 0$

$x^2 - 10x + 9x - 90 = 0$

$x(x-10) + 9(x-10) = 0$

$(x+9)(x-10) = 0$

$\therefore x = -9$ and 10

Hence total number of persons in the function is 10.

OR

Show that the number of diagonals in a polygon of *n* sides is $\frac{n(n-3)}{2}$.

Ans: Let us consider some polygons to prove the given formula.

1. Triangle has 3 sides but 0 diagonals, now let's see if the given formula holds good in this case.
$$\frac{n(n-3)}{2}=\frac{3(3-3)}{2}=0 \text{ true}$$

2. A quadrilateral has 4 sides and 2 diagonals, now let's see if the given formula holds good in this case.
$$\frac{n(n-3)}{2}=\frac{4(4-3)}{2}=2 \text{ true}$$

3. A hexagon has 6 sides and 9 diagonals, now let's see if the given formula holds good in this case.
$$\frac{n(n-3)}{2}=\frac{6(6-3)}{2}=9 \text{ true}$$

42. Calculate the coefficient of variation (C.V.) of the following data:
40, 36, 64, 48, 52.

Ans:

x	d (x-mean)	d^2
40	-8	64
36	-12	144
64	16	256
48	0	0
52	4	16
Total	0	480

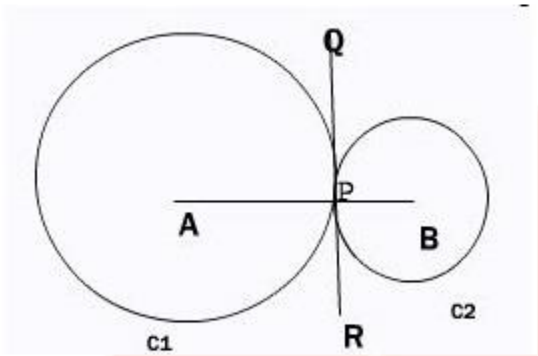
Mean of the given data is 48.

Standard deviation of the given data = $\sqrt{\frac{\sum d^2}{n}}=\sqrt{\frac{480}{5}}=9.8$

Coefficient of Variation = $\frac{\sigma}{\overline{X}}\times 100=\frac{9.8}{48}\times 100=20.42\%$

43. If two circles touch each other externally, prove that the centres and the point of contact are collinear.

Ans:

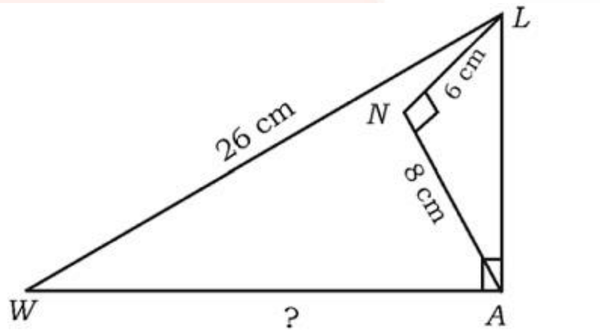


$\angle APQ = \angle BPQ = 90°$ (RQ is the tangent to circles at point P and AP and BP are radii)

$\Rightarrow \angle APQ + \angle BPQ = 180°$

Therefore, the points A, P and B are collinear.

44. In ΔLAW, $\angle$LAW = 90°, $\angle$LNA = 90°, LW = 26 cm, LN = 6 cm and AN = 8 cm. Calculate the length of WA.



Ans: Given that $\angle$LNA = 90°, so ΔANL is a right triangle. Therefore,

$$LA = \sqrt{LN^2 + AN^2}$$
$$LA = \sqrt{6^2 + 8^2}$$
$$LA = \sqrt{36 + 64}$$
$$LA = \sqrt{100} = 10$$

Given that $\angle$LAW = 90°, so ΔLAW is a right triangle. Therefore,

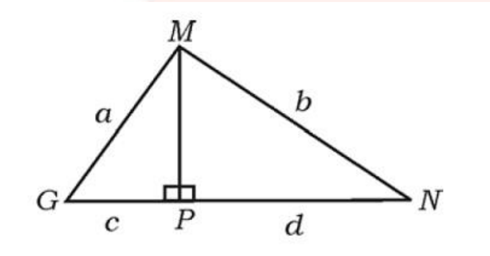
$\sqrt{LW^2 - AL^2} = AW$

$\sqrt{26^2 - 10^2} = AW$

$AW = \sqrt{576} = 24cm$

OR

In Δ MGN, MP⊥ GN. If MG = a units, MN = b units, GP = c units and PN = d units, prove that $\frac{a-b}{c-d} = \frac{c+d}{a+b}$.



Ans. In Δ MGN, MP⊥ GN.
MG = a units,
MN = b units,
GP = c units and
PN = d

In right angled Δ MNP,

$MN^2 = MP^2 + PN^2$

$\Rightarrow MP^2 = MN^2 - PN^2$

$\Rightarrow MP^2 = b^2 - d^2 ....(i)$

In right angled Δ MGP,

$MG^2 = MP^2 + PG^2$

$\Rightarrow MP^2 = MG^2 - PG^2$

$\Rightarrow MP^2 = a^2 - c^2 ....(ii)$

From (i) and (ii), we get

$b^2 - d^2 = a^2 - c^2$

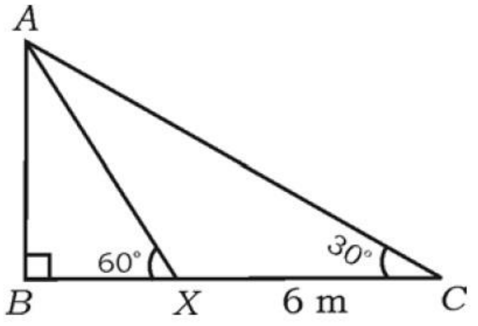
$\Rightarrow a^2 - b^2 = c^2 - d^2$

$\Rightarrow (a+b)(a-b) = (c-d)(c+d)$

$\therefore \frac{(a-b)}{(c-d)} = \frac{(c+d)}{(a+b)}$

Hence proved.

45. The angle of elevation of the top of a flag post (AB) from a point (C) on a horizontal ground is found to be 30°. On walking 6 m towards the post at *X*, the angle of elevation is found to be 60° as shown in the figure. Find the height of the flag post.



Ans:

In ΔABC

$$Tan30° = \frac{AB}{BC}$$

$$\frac{1}{\sqrt{3}} = \frac{AB}{BX+6}$$

$$AB = \frac{BX+6}{\sqrt{3}} \text{..........................(1)}$$

In ΔABX

$$Tan60° = \frac{AB}{BX}$$

$$\sqrt{3} = \frac{AB}{BX}$$

$$AB = \sqrt{3}BX \text{..........................(2)}$$

From (1) and (2)

$$\frac{BX+6}{\sqrt{3}} = \sqrt{3}BX$$

$$BX + 6 = 3BX$$

$$6 = 2BX$$

$$BX = 3m$$

Therefore, $BC = 9m$

From (2)

$AB = \sqrt{3} \times 3 = 3\sqrt{3}m$

OR

Prove that $\dfrac{\sin(90^0 - \theta)}{\cos ec(90^0 - \theta) - \cot(90^0 - \theta)} = 1 + \sin\theta$

Ans:

Ans. L.H.S = $\dfrac{\sin(90^0 - \theta)}{\cos ec(90^0 - \theta) - \cot(90^0 - \theta)}$

$= \dfrac{\cos\theta}{\sec\theta - \tan\theta}$

$= \dfrac{\cos\theta}{\dfrac{1}{\cos\theta} - \dfrac{\sin\theta}{\cos\theta}}$

$= \dfrac{\cos\theta}{\dfrac{1 - \sin\theta}{\cos\theta}}$

$= \dfrac{\cos^2\theta}{1 - \sin\theta}$

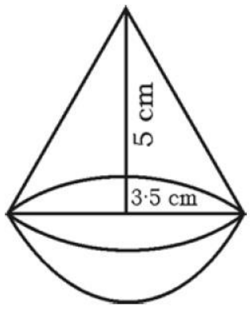
$= \dfrac{1 - \sin^2\theta}{1 - \sin\theta}$ $[\because \sin^2\theta + \cos^2\theta = 1]$

$= \dfrac{(1 - \sin\theta)(1 + \sin\theta)}{1 - \sin\theta}$

$= 1 + \sin\theta$

= R.H.S proved.

46. A toy is in the form of a cone mounted on a hemisphere as shown in the figure. If the radius of each of these solids is $\frac{7}{2}$ cm and height of the cone is 5 cm, find the volume of the toy.



Ans:

Volume of toy = Volume of Cone + Volume of hemisphere

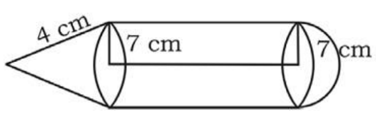
Volume of Cone = $\frac{1}{3}\pi r^2 h = \frac{1}{3}\times\frac{22}{7}\times 3.5\times 3.5\times 5 = 64.17$

Volume of hemisphere = $\frac{4}{3}\pi r^2 = \frac{4}{3}\times\frac{22}{7}\times 3.5\times 3.5 = 51.33$

Volume of toy = $64.17 + 51.33 = 115.5$

OR

A solid is composed of a cylinder with a hemisphere at one end and a cone at other end as shown in the figure. If the radius of each of these solids is7 cm and height of the cylinder is equal to slant height of the cone, find the total surface area of the solid if slant height is 4 cm.



Ans:

Total Surface Area of Solid = Surface Area of cone + Surface Area of Cylinder + Surface area of hemisphere.

Surface Area of cone = $\pi r(r + l) = \frac{22}{7}\times 7(7 + 4) = 242$

Surface Area of Cylinder = $2\pi r(h+r) = 2 \times \frac{22}{7} \times 7(4+7) = 484$

Surface area of hemisphere = $2\pi r^2 = 2 \times \frac{22}{7} \times 7 \times 7 = 308$

Total Surface Area of Solid $= 242 + 484 + 308 = 1034 sq.cm$

V. Answer the following:

47. Construct two circles of radii 4 cm and 2 cm whose centres are 8 cm apart. Construct a transverse common tangent. Measure and write its length. [4]

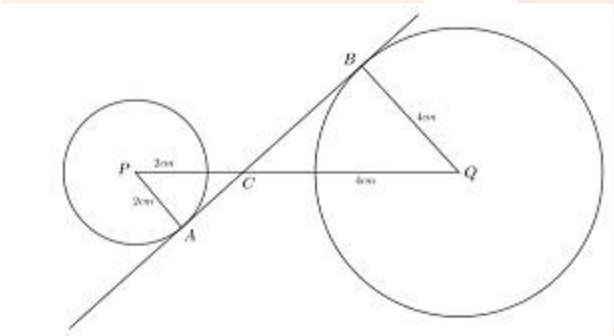
Ans:

Draw a line segment PQ of length 8cm.

With P as a centre draw a circle of radius 2cm and with Q as centre draw a circle with radius 4cm.

Let A and B be the points where the transverse tangent touches these two circles.

Let C be the intersection point of AB and PQ.



In ΔAPC and ΔBQC

$\angle PAC = \angle QBC = 90^\circ$ (Tangent radius property)

$\angle PCA = \angle QCB$ (vertically opposite angles)

$\therefore \Delta APC \sim \Delta BQC$ (AA test)

$$\Rightarrow \frac{PC}{QC} = \frac{PA}{QB} = \frac{2}{4} = \frac{1}{2}$$

$\Rightarrow 2PC = QC$

Given that: $PC + QC = 8$

$$PC + QC = 8$$
$$PC + 2PC = 8$$
$$3PC = 8$$
$$PC = \frac{8}{3}$$
$$QC = \frac{16}{3}$$

$$AC = \sqrt{PA^2 + PC^2}$$
$$AC = \sqrt{4 + \frac{64}{9}}$$
$$AC = \frac{10}{3}$$

Similarly,

$$BC = \sqrt{BQ^2 + QC^2}$$
$$BC = \sqrt{16 + \frac{256}{9}}$$
$$BC = \frac{20}{3}$$

Length of AB = AC + BC = $\frac{10}{3} + \frac{20}{3} = 10cm$

48. State and prove Basic proportionality (Thales) theorem.

Ans:

Basic Proportionality Theorem states that, if a line is parallel to a side of a triangle which intersects the other sides into two distinct points, then the line divides those sides of the triangle in proportion.

In the given figure, if we consider DE is parallel to BC, then according to the theorem,

AD/BD=AE/CE

Proof:

$$Area\ of\ Triangle\ =\ ½\ \times\ base \times\ height$$

In ΔADE and ΔBDE,

$$\frac{Ar\left(ADE\right)}{Ar\left(DBE\right)}=\frac{\frac{1}{2}\times AD\times EF}{\frac{1}{2}\times DB\times EF}=\frac{AD}{DB}..............................\left(1\right)$$

In ΔADE and ΔCDE,

$$\frac{Ar\left(ADE\right)}{Ar\left(ECD\right)}=\frac{\frac{1}{2}\times AE\times DG}{\frac{1}{2}\times EC\times DG}=\frac{AE}{EC}...............................\left(2\right)$$

ΔDBE and ΔECD have a common base DE and lie between the same parallels DE and BC. The triangles having the same base and lying between the same parallels are equal in area.

So, we can say that

$$Ar\left(\Delta DBE\right)=Ar\left(\Delta ECD\right)$$

Therefore,

$$\frac{A\left(\Delta ADE\right)}{A\left(\Delta BDE\right)}=\frac{A\left(\Delta ADE\right)}{A\left(\Delta CDE\right)}$$

Therefore, from (1) and (2)

$$\frac{AD}{BD}=\frac{AE}{CE}$$

Hence Proved.

49. The third term of a geometric progression is square of its first term and the fifth term of it is 64. Find the sum of the first six terms of the Geometric Progression.

Ans: The first term of GP is = a

The third term of GP is = ar^2

$$\Rightarrow ar^2=a^2$$

$$r^2=a$$

The fifth term of GP is = ar^4

$\Rightarrow ar^4 = 64$
$r^6 = 64$
$r = 2$
$\Rightarrow a = 4$

Therefore, the first 6 terms of GP are - $4, 8, 16, 32, 64, 128$ and their sum is 252.

OR

The fourth term of an Arithmetic Progression is 10 and the eleventh term of it exceeds three times the fourth term by 1. Find the sum of the first 20 terms of the progression.

Ans: The first term of AP is = a

The fourth term of AP is = a + 3d

$\Rightarrow a + 3d = 10$..........................(1)

The eleventh term of AP is = a + 10d

$\Rightarrow a + 10d = 3(10) + 1$
$a + 10d = 31$................................(2)

Subtracting (1) from (2)

$7d = 21$
$d = 3$
$\Rightarrow a = 1$

Sum of first 20 terms:

$$S_{20} = \frac{n}{2}\left[2a + (n-1)d\right]$$
$$= \frac{20}{2}\left[2(1) + (20-1)3\right]$$
$$= 10(2 + 57)$$
$$= 590$$

50. Solve the quadratic equation $x^2 - x - 2 = 0$ graphically.

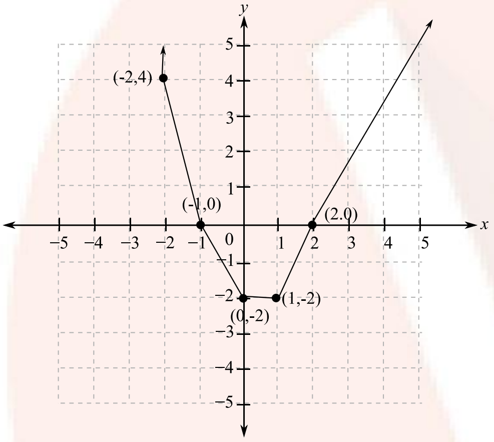
Ans:

$y = x^2 - x - 2$

Now for different values of x, we will find the values of y

x	1	2	0	-1	-2
y	-2	0	-2	0	4

Now we will plot it on graph as below:



Therefore the solution points are -1 *and* 2